

Compound Interest

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What is compound interest?

The basic principle of compound interest is earning **additional interest on interest.**

Once you earn your **first** interest payment, it is **added** to the principle.



Simon has invested \$50,000 for 6 years with Star Bank. The interest rate is 4% pa, compounded annually.

- What would the investment at the end of one year, two years, three years, six years respectively?
- What is the percentage increase on Simon's original investment at the end of year 6?
- What is the total interest at the end of year 6?
- What if the bank paid simple interest? What would be the amount of the interest in this case after year 6?
- Is Simon better off with compound or simple interest?
- What can you conclude? Can you think of a formula to use to calculate compound interest?

After 6 year, Simon's investment is worth
\$ 63 265.95 (Compound interest)

$$\% \text{ increase} = \frac{63265.95 - 50000}{50000} \times 100\% = 26.53\%$$

$$\begin{aligned} \text{Interest earned} &= \$63265.95 - \$50000 \\ &= \$13265.95 \end{aligned}$$

SIMPLE Interest

$$\begin{aligned} I &= PRT \\ &= (50000)(4\%)(6) \\ &= \$12000 \end{aligned}$$

$$\text{Total investment} = \$62000$$

Compound Interest - annual

$$A = P*(1 + R)^n$$

Where

A = final amount

P = principal

R = rate (%) pa

n = number of years

Example – loan (compound interest)

Cathy borrowed \$40,000 from the bank to start her own business. The interest rate is 6% compounded annually. Cathy will pay the whole loan back at the end of the loan period which is 4 years.

- How much would Cathy owe the bank after 2 years?
- What is the total amount Cathy has to pay back the bank?
- What is the total interest paid by Cathy?
- How much would Cathy have to pay the bank if the bank were 6% simple interest?

Compound Interest – periods other than annual

$$\mathbf{A = P * (1 + R)^n}$$

Where

A = final amount

P = principal

R = rate (%)

n = number of compounding periods

Consider the following

- If \$12,000 is invested at 8% per annum interest rate, with interest compounded annually, what is the investment worth after 12 years?
- What if the interest is compounded every 6 months? What is the investment after 12 years?
- What if the interest is compounded every 3 months? What is the investment after 12 years?

$$P = \$12000$$

$$R = 8\% \text{ p.a.}$$

$$n = 12 \text{ years}$$

Compounded annually

$$\begin{aligned} A &= P(1 + 8\%)^{12} \\ &= 12000(1.08)^{12} \\ &= \$30218.04 \end{aligned}$$

Compounded **6**
monthly

$$\text{interest rate} = \frac{8\%}{2}$$

(**two** 6-monthly period)
 $n = 12 \times 2$

$$\begin{aligned} A &= 12000\left(1 + \frac{8\%}{2}\right)^{12 \times 2} \\ &= 12000(1 + 4\%)^{24} \\ &= 12000(1.04)^{24} \\ &= \$30759.65 \end{aligned}$$

Compounded **3**
monthly
interest rate = $\frac{8\%}{4}$

(**Four** 3-monthly
period,
 $n = 12 \times 4 = 48$)

$$\begin{aligned} A &= 12000(1 + 2\%)^{48} \\ &= \$31044.84 \end{aligned}$$
